Discrete Fourier Transform (DFT)

Definition

DFT is a mathematical operation that converts a finite sequence of discrete samples into a sequence of frequencies. In simple terms, DFT is a way to break something complicated into simple repeating terms.

DFT Formula

The discrete Fourier transform of a sequence of N points is given as:

$$\Phi_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} a_j\, e^{2\pi ijk/N} \qquad k = 0, 1, 2, \ldots, N-1 \tag{1}$$

Using Euler's formula $e^{ix} = \cos x + i \sin x$:

$$\Phi_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} a_j \left[\cos\left(\frac{2\pi jk}{N}\right) + i \sin\left(\frac{2\pi jk}{N}\right)\right] \tag{2}$$

Example 1

Given: $a_0 = 0.841,\ a_1 = 0.909,\ a_2 = 0.141,\ a_3 = -0.757$. Find DFT ($\Phi_0$, Φ_1, Φ_2) with $N = 4$.

For $N = 4$ the formula reduces to:

$$\Phi_k = \frac{1}{2} \sum_{j=0}^{3} a_j\, e^{\pi ijk/2}$$

$k = 0$

$$\Phi_0 = \frac{1}{2}[a_0 + a_1 + a_2 + a_3] = \frac{1}{2}[0.841 + 0.909 + 0.141 - 0.757]$$

$$\boxed{\Phi_0 = 0.567}$$

$k = 1$

$$\Phi_1 = \frac{1}{2}\left[a_0 + a_1 e^{\pi i/2} + a_2 e^{\pi i} + a_3 e^{3\pi i/2}\right]$$

$$= \frac{1}{2}\left[0.841 + 0.909\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) + 0.141(\cos\pi + i\sin\pi) - 0.757\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)\right]$$

$$= \frac{1}{2}\left[0.841 + 0.909i + 0.141(-1) - 0.757(-i)\right] = \frac{1}{2}\left[0.982 + 1.666i\right]$$

$$\boxed{\Phi_1 = 0.491 + 0.833\, i}$$

$k = 2$

$$\Phi_2 = \frac{1}{2}\left[a_0 + a_1 e^{\pi i} + a_2 e^{2\pi i} + a_3 e^{3\pi i}\right]$$

$$= \frac{1}{2}[0.841 + 0.909(-1) + 0.141(1) - 0.757(-1)] = \frac{1}{2}[0.841 - 0.909 + 0.141 + 0.757] = \frac{1}{2}[0.83]$$

$$\boxed{\Phi_2 = 0.415}$$

Result: The discrete Fourier transform values are $\Phi_0 = 0.567, \quad \Phi_1 = 0.491 + 0.833\,i, \quad \Phi_2 = 0.415.$

Example 2

Given: $a_0 = 1,\ a_1 = 0,\ a_2 = -1,\ a_3 = 0$. Find DFT with $N = 4$.

$k = 0$

$$\Phi_0 = \frac{1}{2}[1 + 0 - 1 + 0] = 0 \qquad \boxed{\Phi_0 = 0}$$

$k = 1$

$$\Phi_1 = \frac{1}{2}\left[1 + 0 + (-1)e^{\pi i} + 0\right] = \frac{1}{2}[1 - (-1)] = \frac{1}{2}(2) \qquad \boxed{\Phi_1 = 1}$$

$k = 2$

$$\Phi_2 = \frac{1}{2}\left[1 + 0 + (-1)e^{2\pi i} + 0\right] = \frac{1}{2}[1 - 1] = 0 \qquad \boxed{\Phi_2 = 0}$$

Example 3

Given: $a_0 = 2,\ a_1 = 0,\ a_2 = -2,\ a_3 = 0$. Find DFT with $N = 4$.

$k = 0$

$$\Phi_0 = \frac{1}{2}[2 + 0 - 2 + 0] = 0 \qquad \boxed{\Phi_0 = 0}$$

$k = 1$

$$\Phi_1 = \frac{1}{2}\left[2 - 2\left(\cos\pi + i\sin\pi\right)\right] = \frac{1}{2}[2 - 2(-1)] = \frac{1}{2}(4) \qquad \boxed{\Phi_1 = 2}$$

$k = 2$

$$\Phi_2 = \frac{1}{2}[2 - 2] = 0 \qquad \boxed{\Phi_2 = 0}$$

DFT for a 3-Qubit System

For a 3-qubit system: $N = 2^3 = 8$, so $k = 0, 1, 2, 3, 4, 5, 6, 7$.

The DFT in matrix form:

$$\begin{pmatrix} \Phi_0 \\ \Phi_1 \\ \vdots \\ \Phi_{N-1} \end{pmatrix} = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & & & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)^2} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{pmatrix}$$

where $\omega = e^{2\pi i/N}$.

(i) $|000\rangle$: $a = [1, 0, 0, 0, 0, 0, 0, 0]$

Since $a_0 = 1$ and all others are 0:

$$\Phi_k = \frac{1}{\sqrt{8}} \cdot 1 = 0.3536 \quad \text{for all } k$$

$$\Phi_k = [0.3536,\ 0.3536,\ 0.3536,\ 0.3536,\ 0.3536,\ 0.3536,\ 0.3536,\ 0.3536]^T$$

(ii) $|001\rangle$: $a = [0, 1, 0, 0, 0, 0, 0, 0]$

Only $a_1 = 1$, so $\Phi_k = \dfrac{\omega^k}{\sqrt{8}}$ where $\omega = e^{\pi i/4}$.

By Euler's formula:

$$\Phi_k = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 \\ e^{\pi i/4} \\ e^{\pi i/2} \\ e^{3\pi i/4} \\ e^{\pi i} \\ e^{5\pi i/4} \\ e^{3\pi i/2} \\ e^{7\pi i/4} \end{pmatrix} = \begin{pmatrix} 0.3536 \\ 0.25 + 0.25i \\ 0 + 0.3536i \\ -0.25 + 0.25i \\ -0.3536 \\ -0.25 - 0.25i \\ 0 - 0.3536i \\ 0.25 - 0.25i \end{pmatrix}$$

(iii) $|010\rangle$: $a = [0, 0, 1, 0, 0, 0, 0, 0]$

Only $a_2 = 1$, so $\Phi_k = \dfrac{\omega^{2k}}{\sqrt{8}}$.

$$\Phi_k = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 \\ e^{\pi i/2} \\ e^{\pi i} \\ e^{3\pi i/2} \\ e^{2\pi i} \\ e^{5\pi i/2} \\ e^{3\pi i} \\ e^{7\pi i/2} \end{pmatrix} = \begin{pmatrix} 0.3536 \\ 0 + 0.3536i \\ -0.3536 \\ 0 - 0.3536i \\ 0.3536 \\ 0 + 0.3536i \\ -0.3536 \\ 0 - 0.3536i \end{pmatrix}$$